

Cloud Computing Practical - Launch Virtual Machine using AWS EC2

Aim: To study and implement the procedure for launching a Virtual Machine using Amazon Web Services EC2.

Theory:

Amazon EC2 (Elastic Compute Cloud) is a cloud computing service provided by AWS that allows users to launch virtual servers in the cloud. EC2 provides scalable computing resources, flexible instance types, storage options and networking facilities. A virtual machine launched in AWS is called an EC2 Instance.

Requirements:

1. AWS Account
2. Stable Internet Connection
3. Web Browser
4. SSH Client like Git Bash or PuTTY

Detailed Procedure

Step 1: Open browser and visit AWS Management Console.

Step 2: Login using AWS username and password.

Step 3: In AWS Search Bar, search for EC2.

Step 4: Open EC2 Dashboard.

Step 5: Click on Launch Instance button.

Step 6: Enter Instance Name such as MyEC2Instance.

Step 7: Select Operating System Image such as Ubuntu Server 22.04.

Step 8: Select Instance Type like t2.micro which is available in free tier.

Step 9: Click Create New Key Pair.

Step 10: Enter key pair name and download PEM file.

Step 11: Under Network Settings allow SSH traffic from Anywhere.

Step 12: Configure Storage according to requirement.

Step 13: Review all configurations carefully.

Step 14: Click Launch Instance.

Step 15: Wait for instance state to change from Pending to Running.

Step 16: Select the instance and copy Public IPv4 address.

Step 17: Open terminal or Git Bash.

Step 18: Navigate to folder where PEM file is downloaded.

Step 19: Connect to EC2 instance using SSH command.

Step 20: After successful login, execute Linux commands.

SSH Command

```
chmod 400 mykey.pem  
ssh -i mykey.pem ubuntu@Public-IP-Address
```

Basic Linux Commands Executed on EC2

```
sudo apt update  
sudo apt upgrade -y  
mkdir project  
cd project  
touch demo.txt  
ls  
pwd
```

Advantages of AWS EC2

1. Scalable virtual machines
2. Pay only for used resources
3. Highly secure infrastructure
4. Easy remote access using SSH
5. Supports multiple operating systems

Explanation:

AWS EC2 provides Infrastructure as a Service (IaaS) where users can launch virtual machines in cloud environment. Users can choose operating system, RAM, CPU and storage according to application requirements. The virtual machine can be connected remotely using SSH protocol. AWS automatically manages physical hardware, networking and availability.

Result: Virtual Machine was launched and accessed successfully using AWS EC2 cloud platform.